1. Two researchers, A and B, perform a three-stage Stern–Gerlach experiment. In each stage, they allow only the spin-up (+) component to pass to the next magnet.

- Researcher A uses magnetic fields in the sequence: $\hat{z} \to (-\hat{z} + \sqrt{3}\hat{y})/2 \to \hat{z}$
- Researcher B uses magnetic fields in the sequence: $\hat{z} \to \hat{x} \to \hat{z}$

If both start with identical electron fluxes, estimate the ratio of the final number of spin-up electrons measured by A compared to B. [5]

2. Fundamental forces are mediated by the exchange of gauge bosons.

(a) List the gauge bosons responsible for the Strong, Electromagnetic, and Weak interactions.

(b) Identify one fermion from the Standard Model that does not experience the strong interaction and one that does not experience the electromagnetic interaction.

Gluons (g°)

[3 + 2]

3. Suppose a beam of electrons can reach a detector via two paths with amplitudes $\alpha_1 = 1 + i$ and $\alpha_2 = -1 + i$. Calculate the probability of detecting an electron at the specific point where these two paths meet if

(a) the electron is not watched, and

(b) a light source is turned on to watch the electrons.

[5]

4. Consider a particle trapped in a one-dimensional box of length a centered at the origin. At an instant t, the probability density for a non-stationary state is given by

$$|\Psi(x)|^2 = \frac{1}{a}\left[1 + \sin\left(\frac{2\pi x}{a}\right)\right], \quad \text{for} \quad -a/2 \le x \le a/2$$

(a) Calculate the expectation value of the position $\langle x \rangle$.

(b) Show whether the particle is more likely to be found in the region $x > 0$ or $x < 0$.

[2.5 + 2.5]

5. Relativistic dynamics gives $E^2 = p^2c^2 + m^2c^4$.

(a) For a particle of mass m, determine the ratio of momentum to total energy as its velocity approaches the speed of light ($v \to c$). Express your result in terms of γ, β (standard notation).

(b) Using the definition of force $\mathbf{F} = \frac{d\mathbf{p}}{dt}$, show that maintaining a constant acceleration for a massive particle requires a force that increases as a function of velocity. Use this to explain the physical limit on reaching $v = c$.

[2 + 3]

$2 \quad p^2c^2 + m^2c^4$

6. Show that the Schrödinger equation retains the normalisation of its solutions.

7. What is cosmological principle and at what length scale is it valid? [2+1]

cosmological principle states that at large enough scale

8. (a) State Hubble's law.

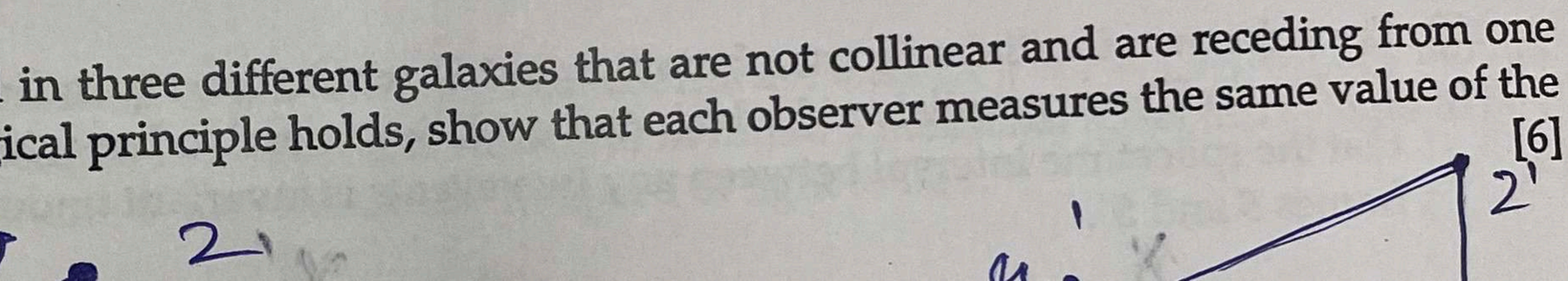
(b) Using Hubble's law, estimate the distance from Earth to a galaxy whose recession speed equals the speed of light. You may use

$$c = 3 \times 10^8 \text{ m s}^{-1}, \qquad H_0 = 72 \text{ km s}^{-1} \text{ Mpc}^{-1},$$

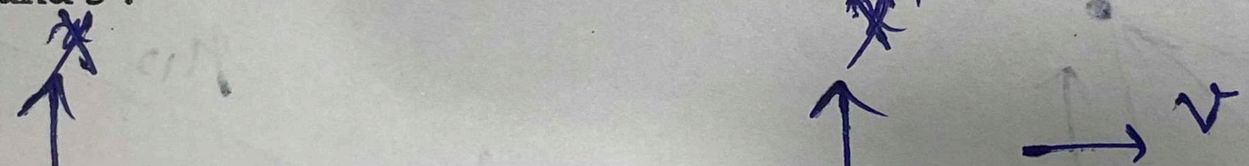
and 1 pc = 3.26 light-years.

(c) Can galaxies recede from us with speeds greater than the speed of light? Briefly explain. [2+5+1]

9. Consider three observers located in three different galaxies that are not collinear and are receding from one another. Assuming the cosmological principle holds, show that each observer measures the same value of the Hubble constant. [6]



10. Consider two inertial reference frames S and S' whose corresponding coordinate axes (*i.e.*, X with X', Y with Y', Z with Z') are parallel, and which move relative to each other with velocity v along the Y (or Y') axis. Prove that the spacetime interval between any two events is invariant under the Lorentz transformations relating the frames S and S'. [7]



11. Consider two inertial reference frames S and S' with parallel coordinate axes ($X \parallel X'$, $Y \parallel Y'$, $Z \parallel Z'$), moving with relative velocity v along the common Z-axis. If an observer in frame S measures the speed of light to be c, demonstrate that an observer in frame S' also measures the speed of light to be c. [6]